

as the downconverting medium, which efficiently absorbs blue light emitted by, for example, highly efficient gallium nitride LED and converts it into another wavelength of choice.

One of the most serious obstacles to using quantum-dot LEDs in consumer products is the toxicity of cadmium- and lead-based quantum dots. This problem has been addressed by developing high-luminescence, non-toxic quantum dots using InP/ZnS core shell.

First, nanoparticles of quantum dots using InP/ZnS with tunable

size and colour are made. It is then followed by a wet chemical process to form a uniform inorganic protective shell on the quantum dots to achieve stability of the quantum dots and to improve the reliability of the polymer encapsulation of the LEDs. Then the encapsulated quantum dots with shells are further modified to make them well dispersed in the polymer matrix. The luminescent, spectrally-tunable quantum dots are then coupled with single-crystal LED chips to serve as a colour converter. Fig. 5 shows the scheme of a non-toxic quantum-dot LED.

3 Building materials

In a typical office building, air-conditioning is the largest energy guzzler, which consumes about 48 per cent of the total energy. So the demand for better thermal insulation in the building enclosures has dramatically increased over the last few years.

To improve thermal insulation of composite wall panels, it is not practical to increase the wall thickness significantly. Research in this area is therefore focused on reducing the thermal conductivity of the wall materials without compromising on the other beneficial properties.

Introduction of porosity is the most straightforward way of improving thermal insulation. But porosity reduces structural strength and increases moisture uptake, which leads to corrosion. Nanomaterials and nanotechnology can address this issue. One solution is to use thermally insulating coating of aerogels and titanium dioxide (TiO_2) nanoparticles on building materials like conventional glass, tiles and aluminium plates.

The high porosity of aerogels makes these an excellent thermal insulator. However, when radiation becomes a significant heat transfer mechanism at high temperature, the insulative performance of aerogels decreases since these are highly transparent in the 3-8 μm wavelength region. With the addition of opacifiers, e.g.,

TiO_2 , this infrared transparency of aerogels can be reduced or even eliminated, ensuring that the insulative coating is suitable for a wide range of climates.

Another merit of nano TiO_2 particles is their photo-induced superhydrophilicity property that can provide a self-cleaning function. This property makes these particles one of the most promising contenders for insulative coating in outdoor applications.

Another solution to reduce electricity consumption is use of high-performance cementitious materials for construction of the external wall with enhanced thermal insulation. For a given wall thickness, the thermal conductivity could be reduced if solid concrete is replaced with foamed concrete.

To meet the structural requirement, the foamed concrete should have compression strength equal to or greater than 30 MPa. For reducing the electrical consumption of the air-conditioning system, it should have thermal insulation property of less than 1W/mK. To achieve these technical requirements, lightweight sand and a foaming agent are used to make lightweight concrete with low thermal conductivity.

To maximise the strength, Pozzolans along with thermal curing is provided. Further, to prevent corrosion of steel reinforcement, a fibre-reinforced cementitious composite composition is applied to the wall surface. It acts as the barrier to water and chemical penetration. It has been found that by suitably adopting this technology electric power consumption of air-conditioning can be reduced by half.

4 Carbon-nanotube GHz amplifier

Carbon nanotube field-effect transistors (CNFETs) have been used recently to build a GHz amplifier. CNFETs are very attractive for future RF applications such as amplifiers, mixers and switches. Carbon nanotubes and graphene materials have superior electrical, mechanical strength and thermal properties, thanks to their one-dimensional transport and extremely strong carbon-carbon bonds leading to large mean free paths, high current densities and low thermal noise. Researchers are investigating ways of exploiting these characteristics to develop better performing electronic devices.

Ballistic transport of electrons through semiconducting carbon nanotubes has led to the development of CNFETs—one of the key building blocks of any electronic application. The one-dimensional tube structure allows quantum-capacitance limited operation, resulting in low input and feedback capacitance. It also significantly reduces the scattering probability, thus promising high-frequency performance with low figures. The structure also causes the drain current to be linearly dependent on the applied gate voltage, promising uniquely linear devices.

CNFETs are thermally robust, which simplifies the heat management. The current conventional semiconductor devices used in RF front-ends can be designed with high linearity but at the expense of reducing the operating efficiency to about 5 per cent.

CNFETs recently developed deliver similar linearity but at 70 per cent less dissipated power. The system has deep impact on the battery life and reduction in heat dissipation.

Typical construction steps are as follows: In a CNFET, thousands of carbon nanotubes are laid out between the drain and the source, providing a low resistance channel as shown in Fig. 6. The current flow through the channel is controlled by a top gate, separated from the tube array by a high-K dielectric. The base substrate is Si, which serves as the mechanical platform. A thick SiO_2 is grown over it to

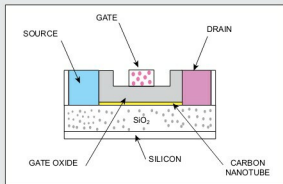


Fig. 6: Cross-section of a CNFET